

# To Spend a Stone with Six Birds: Currency–Constraint Duality and Shadow Prices Across Closure Layers

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## Abstract

Physics has temperatures, economics has prices, statistics has regularization strengths, and models of attention have precision weights. Each is a single number that prices a limited resource. This paper asks why such numbers appear whenever a stable descriptive layer forms, and answers within the Six Birds closure calculus. We define a currency relative to a layer by the job it does: bounding what is feasible, adding up along protocols, driving a monotone potential, or pricing a constraint. We then state the currency–constraint principle as a conditional result. If maintaining a higher-level object requires keeping a lower-level ledger within a budget, that ledger becomes a feasibility constraint for the higher layer; if the higher layer then closes by maximizing entropy under the budget, the Gibbs multiplier of the constraint is a nonnegative shadow price. We prove this finite duality, including exactly when the price is positive, zero, or infinite. We also formalize in Lean 4 the single-row duality and the data-processing inequality that prevents coarse observation from creating path-reversal asymmetry. A finite Markov laboratory illustrates each step. Coarse observation never inflates a regularized path audit. The fitted price follows the predicted curve and vanishes once the budget is slack. A resolution ladder exposes more cycle directions (0, 5, 11) than are actually driven (0, 0, 6). A ledger-derived cost predicts held-out transitions better than a scale-matched proxy. Stronger maintenance pricing shrinks packaging defects under an explicit bound. Persistence alone does not create a budget; the principle applies once a resource law is given, and prices then follow whenever the layer closes by maximum entropy under a feasible budget.

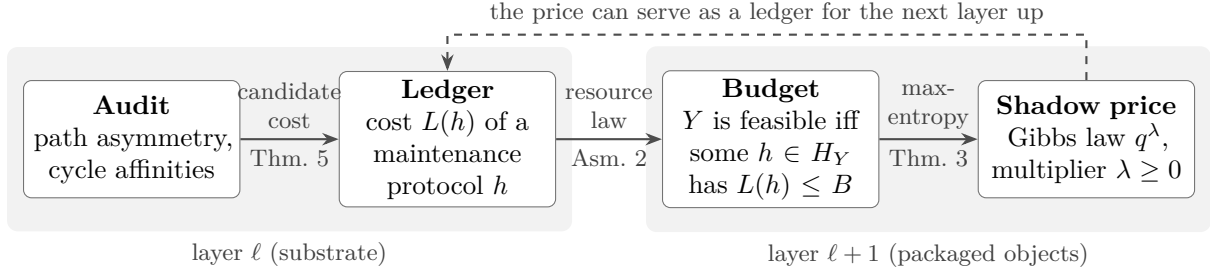
## 1 Introduction

### 1.1 Currencies keep reappearing

Energy in physics, code length in information theory, money in economics, compute in algorithms, and attention in cognition are introduced in different languages, yet they do the same mathematical work. They limit which states or protocols are feasible. They add up along sequences of operations. And they come back as exchange rates between goals that cannot all be met at once: inverse temperature trades entropy against energy, a market price trades one good against another, a regularization strength trades fit against complexity [1, 8, 17, 18]. We call such quantities *currencies*.

Standard duality theory explains part of this [3]. Once an optimization problem is written down, with its variables and its constraints, Lagrange multipliers are exactly what one should expect. But that explanation starts late. It takes the descriptive layer and its constraints as given. It does not say where a constraint at one level of description comes from, or why it so often looks like a limit on something spent at a lower level.

This paper supplies that missing step, and is careful about what the step requires. The idea is simple. A higher-level object, such as a macrostate, a packaged pattern, or a stable representation, persists only if something at the lower level keeps maintaining it. When that maintenance draws on a limited resource, the resource bound becomes a feasibility constraint at the higher level. If the higher layer then settles into its least-committed (maximum-entropy)



**Figure 1:** The currency–constraint chain. A ledger at layer  $\ell$  records what maintenance costs. Audits measure irreversible drive; the path-reversal KL audit measures directional asymmetry that coarse observation cannot inflate (Theorem 5). Using an audit quantity as a maintenance ledger is a modelling choice, not a consequence of the theorem. A resource law, supplied as a premise, turns the ledger into a feasibility constraint on objects of layer  $\ell + 1$ . When layer  $\ell + 1$  closes by maximum entropy under that constraint, the Gibbs multiplier is a shadow price, which is a currency of the new layer.

description subject to that constraint, a multiplier appears, and the multiplier is the higher layer’s own currency: the exchange rate between what it would like to do and what the lower layer can afford. Figure 1 shows the chain.

The qualifier matters. Persistence by itself does not imply a budget: a system that stays in one state forever persists perfectly and spends nothing. The principle therefore takes the resource law as a premise and proves what follows from it. Within that scope the conclusions are theorems, not analogies.

## 1.2 The Six Birds setting

Six Birds is a closure calculus for emergent description [21]. It is built from six primitives: **P1** operator rewrite, **P2** constraints or gating, **P3** protocol holonomy (route dependence), **P4** staging or sectors, **P5** packaging, and **P6** accounting or audit. This paper is about the chain **P5**→**P6**→**P2**: packaging creates higher-level objects, accounting measures what it costs to keep them coherent, and constraints turn that cost into the boundary of what the next layer can do.

The framework also tells us what does *not* count. A quantity is a currency only if it does structural work: it gates feasibility, adds along protocols, drives a monotone potential, or prices a constraint. A statistic that merely correlates with behaviour is not enough, and neither is bare route dependence. Protocol mismatch becomes a directionality currency only when an honest audit backs it up, and an audit is honest only if coarse observation cannot manufacture it.

## 1.3 Contributions

1. **A layer-relative definition of currency** (Section 2) with four roles (constraint, ledger, potential, dual price) and the nontriviality conditions that separate them from arbitrary statistics.
2. **A conditional currency–constraint principle** (Section 3). Given a maintenance ledger and a resource law, the ledger induces a higher-layer feasibility constraint. In a finite entropy closure the Gibbs law is optimal and its multiplier is a supporting shadow price (Theorem 3). We describe the full price curve, including infeasible, infinite-price, active, and slack regimes (Proposition 4).
3. **Honest audits** (Section 4). Path-reversal relative entropy cannot increase under any deterministic pointwise observation. A simple regularization keeps empirical audits finite while preserving that inequality on the same sample. Cycle affinities give low-dimensional drive coordinates.
4. **A finite Markov laboratory** (Section 5) that illustrates each piece and separates what is forced by theorem from what depends on the chosen example. This includes a uniform

**Table 1:** What each main claim rests on. “Lean” means machine-checked; “proof” means a complete proof in this paper; “example” means it holds in the stated finite construction and is not claimed in general.

| Claim                                                                                | Support         | Where             |
|--------------------------------------------------------------------------------------|-----------------|-------------------|
| Deterministic observation cannot increase finite supported KL, or path-reversal KL   | Lean            | Thm. 5, App. B    |
| The reversal-mixture audit obeys the same inequality on a single sample              | proof           | Prop. 6           |
| Gibbs optimality and the shadow-price inequality (one row)                           | Lean            | Thm. 3, App. B    |
| Weighted rows; price regimes; $d\lambda^*/db < 0$                                    | proof           | Prop. 4, App. A   |
| Persistence alone does not imply a budget                                            | counterexample  | Example 1         |
| Packaging defect $\leq \min(2, 2\tau\varepsilon)$ ; vanishes as the exit price grows | proof           | Thm. 8            |
| Cycle ranks 0, 5, 11 and drive rank 0, 0, 6 on the ladder                            | proof + example | Prop. 7, Sec. 5.4 |
| Coarsening can create cycle rank                                                     | counterexample  | Sec. 5.4          |
| A fixed kernel other than the truth has positive expected excess held-out loss       | proof           | Sec. 5.5          |
| The fitted ledger model beats a scale-matched proxy on held-out data                 | example         | Sec. 5.5          |

bound showing that stronger maintenance pricing makes packaged objects nearly idempotent (Theorem 8).

5. **A Lean 4 formalization** (Appendix B) of finite supported KL data processing, its path-reversal specialization, and single-row Gibbs duality, using only the standard axioms.

Table 1 lists every main claim and the kind of support behind it. The distinction matters because the laboratory is synthetic. Where a pattern is guaranteed by a theorem, the numerics serve as a consistency check; where it is not, we say so.

#### 1.4 Place in the Six Birds series

*Six Birds: Foundations of Emergence Calculus* introduced the primitives and the layer package of lenses, packaging, and audits [21]. *To Create a Stone with Six Birds* turned them into an interaction algebra (PICA) and a finite stochastic laboratory [20]. The present paper adds the economics of closure: what a closure must spend to keep its objects, how that spending becomes the next layer’s constraint, and when the next layer answers with a price.

## 2 Currencies relative to a layer

### 2.1 Layers as closure packages

In *Six Birds* a descriptive layer is a closure package, not a metaphysical level [21]. It consists of a finite substrate, a lens that fixes what can be said at that resolution, a packaging rule that turns recurrent structure into objects, and an audit that certifies what survives coarse observation. In the notation of the foundations paper,

$$\mathcal{T} = (Z, f, \Sigma_f, E, \mathcal{A}),$$

where  $Z$  is a finite substrate,  $f$  a lens,  $\Sigma_f$  the induced definability algebra,  $E$  a packaging endomap, and  $\mathcal{A}$  an audit functional.

Currencies are always relative to such a package. The same number can play different roles at different resolutions. Energy is a conserved or budgeted quantity for microscopic dynamics; at the ensemble level the natural coordinate is inverse temperature, the multiplier of the mean-energy constraint. A regularization strength is not a currency of the task being learned; it is a currency of the learning layer, which uses it to decide which updates it can afford.

## 2.2 Four structural roles

Fix a layer with state space  $Y$ . A currency is a low-dimensional map  $u : Y \rightarrow \mathbb{R}^k$ , usually with  $k \ll |Y|$ , that plays at least one of the following roles.

**Constraint currency (C1).** Feasibility is expressed through  $u$ : the feasible set is

$$K = \{y \in Y : g(u(y)) \leq 0\}$$

or an intersection of such sets. This is the primal view. The currency is what the layer must keep within bounds.

**Ledger currency (C2).** Costs add along protocols. For a protocol  $\gamma = (y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_T)$ ,

$$\text{Cost}(\gamma) = \sum_{t=0}^{T-1} \ell(u(y_t), y_t \rightarrow y_{t+1}),$$

exactly or in expectation. This is the accounting view.

**Potential currency (C3).** A scalar potential  $U$ , built from the ledger, decreases along trajectories:  $U(y_{t+1}) \leq U(y_t)$  up to controlled noise, or  $\dot{U} \leq 0$  in continuous time. This is the Lyapunov or thermodynamic view.

**Dual currency (C4).** The layer closes by optimizing under constraints, and the currency is a shadow price. For a concave objective  $\Phi$  and an inequality constraint,

$$V(b) = \sup_{q \in \mathcal{Q}} \{\Phi(q) : \mathbb{E}_q[f] \leq b\},$$

a multiplier  $\lambda \geq 0$  prices the constraint. It satisfies  $V(b') \leq V(b) + \lambda(b' - b)$  for all  $b'$ ; that is,  $\lambda$  is a supergradient of the concave value function. Where  $V$  is differentiable this reads  $\lambda = V'(b)$ . An inequality constraint gives a nonnegative price, while an equality constraint gives a multiplier of either sign. A constraint can also be active with price zero; in the finite entropy closure of Section 3 this happens exactly at the threshold  $b = c_0$  where the unconstrained optimum just meets the budget (Proposition 4).

These roles come from standard places. Shadow prices are the dual objects of constrained optimization [3]; money has a well-known interpretation as a ledger or memory device [10]; and thermodynamic potentials arise as constrained duals in the maximum-entropy programme [8].

**Nontriviality.** Read literally, each role can be satisfied trivially. A constant  $u$  with  $K = Y$  meets C1; an edge cost that ignores  $u$  meets C2; a constant potential meets C3. To distinguish structural currencies from arbitrary statistics we therefore require essential use. The feasibility boundary must be proper ( $K \neq Y$ ) and must depend on  $u$ . The ledger increment must depend on  $u$ . The potential must be nonconstant and tied to the ledger. And a dual currency must belong to a constraint that can bind. For the dual role the condition has a precise form. A price is identifiable only if the cost is nonconstant in some row that carries weight (Proposition 4); every cost used in Section 5 has this property.

### 2.3 Structural currencies, proxies, and path quantities

Many quantities called currencies in applied work are proxies. Trust scores, prestige metrics, and ad hoc complexity measures can correlate with feasibility or performance without entering the layer's constraints, ledgers, or dual programmes. They may be useful summaries, but they are not currencies in the sense used here. Section 5.5 makes the distinction concrete. A cost with exactly the same values in each row, but attached to the wrong transitions, predicts worse than the cost read off the lower-layer ledger.

Currencies can live on states or on paths. A state currency is a function  $u(y)$ . A path currency is a functional of whole trajectories: an action, an accumulated entropy production, or a path-reversal asymmetry. Six Birds allows both, because audits can be defined on states, edges, or paths [21]. What matters is the structural role, not the representation.

### 2.4 Why route dependence is not enough

Protocol mismatch, or holonomy (**P3**), is often the first sign that a system remembers how it was driven. But route dependence alone is not a directionality currency. A protocol can fail to commute simply because a schedule variable has been hidden, while the enlarged autonomous system is perfectly reversible. What counts is an audit that detects irreversible drive and that coarse observation cannot inflate. Path-reversal relative entropy is the classical choice: irreversibility is measured by comparing the law of forward paths with the law of reversed paths [11, 12, 16]. Section 4 proves that this audit can be hidden by observation but never created by it, and builds the empirical version used in the laboratory.

## 3 From maintenance ledgers to shadow prices

### 3.1 The bridge from persistence to a budget

Let layer  $\ell$  have states  $y_\ell \in Y_\ell$ , and let a packaging map  $\Pi : Y_\ell \rightarrow Y_{\ell+1}$  define the objects of layer  $\ell + 1$ . One might hope that the mere persistence of a packaged object forces a bound on lower-level spending. It does not.

**Example 1** (Persistence without spending). Take a single state, the identity kernel, and the identity packaging map. The packaged object persists forever and is a fixed point of every packaging rule, yet any ledger that is identically zero is consistent with it. Equally, the same dynamics is consistent with an externally imposed ledger that is unbounded. Persistence alone fixes no resource bound.

A budget must therefore be supplied. We state the premise explicitly.

**Assumption 2** (Maintenance ledger and resource law). Layer  $\ell$  comes with a set of maintenance protocols  $h$ , a nonnegative ledger  $L(h)$ , and a resource bound  $B$ : a protocol is admissible if and only if  $L(h) \leq B$ . Each object  $Y$  of layer  $\ell + 1$  has a set  $H_Y$  of protocols that would maintain it.

Under this assumption, an object is maintainable exactly when some admissible protocol maintains it:

$$Y \in K \iff \exists h \in H_Y \text{ with } L(h) \leq B.$$

The lower-layer ledger has become a feasibility constraint (role C1) at layer  $\ell + 1$ . If the infimum  $u(Y) = \inf_{h \in H_Y} L(h)$  is attained, this is the familiar form  $K = \{Y : u(Y) \leq B\}$ . Over a fixed horizon  $t > 0$  the same statement holds for average cost,  $L(h)/t \leq B/t$ . Two cautions apply. Feasibility is inherited through an existential witness: it says that some maintenance protocol exists, not that every transition written at the higher layer is realizable at the lower one. And the assumption, not persistence, supplies  $B$  and  $H_Y$ .

### 3.2 Finite maximum-entropy closure

Suppose now that layer  $\ell + 1$  chooses a stochastic description by maximizing entropy subject to a budget on expected cost. This is the maximum-caliber principle for transitions [8, 14], and it is the simplest closure in which a price can appear.

Fix finite nonempty sets of rows  $I$  and destinations  $J$ , a real cost array  $u_{ij}$ , and fixed row weights  $\mu_i \geq 0$  with  $\sum_i \mu_i = 1$ . For a row-stochastic array  $q$  (each  $q_{i\cdot}$  a probability vector), define

$$H_\mu(q) = - \sum_{i,j} \mu_i q_{ij} \log q_{ij}, \quad C_\mu(q) = \sum_{i,j} \mu_i q_{ij} u_{ij}.$$

The weights  $\mu$  are fixed in advance. Letting them depend on  $q$ , for instance through the stationary law of  $q$ , would define a different problem. For  $\lambda \in \mathbb{R}$  the Gibbs array is

$$q_{ij}^\lambda = \frac{e^{-\lambda u_{ij}}}{Z_i(\lambda)}, \quad Z_i(\lambda) = \sum_j e^{-\lambda u_{ij}}.$$

**Theorem 3** (Finite Gibbs duality). *Let  $\lambda \geq 0$  and  $b \in \mathbb{R}$ .*

(i) *For every row-stochastic  $p$ ,*

$$\sum_i \mu_i D_{\text{KL}}(p_{i\cdot} \parallel q_{i\cdot}^\lambda) = -H_\mu(p) + \lambda C_\mu(p) + \sum_i \mu_i \log Z_i(\lambda).$$

(ii) *Consequently  $H_\mu(p) - \lambda C_\mu(p) \leq H_\mu(q^\lambda) - \lambda C_\mu(q^\lambda)$  for every  $p$ .*

(iii) *If  $C_\mu(q^\lambda) \leq b$  and  $\lambda(C_\mu(q^\lambda) - b) = 0$ , then  $q^\lambda$  maximizes  $H_\mu$  over  $\{p : C_\mu(p) \leq b\}$ .*

(iv) *Under the conditions of (iii), every  $p$  with  $C_\mu(p) \leq b'$  satisfies  $H_\mu(p) \leq H_\mu(q^\lambda) + \lambda(b' - b)$ . Hence  $\lambda$  is a supergradient of the concave value function  $V(b) = \sup\{H_\mu(p) : C_\mu(p) \leq b\}$  at  $b$ , and  $V'(b) = \lambda$  wherever  $V$  is differentiable.*

*In (iii) the optimizer is unique on the rows with  $\mu_i > 0$ ; rows with  $\mu_i = 0$  do not enter the objective, and there any row is optimal.*

The proof is a short calculation (Appendix A). For a single row, the Lean development constructs the Gibbs law from exponentials, proves that it is normalized and strictly positive, and verifies (i)–(iii) together with the inequality in (iv) (Appendix B). In both versions feasibility and complementary slackness are hypotheses. The theorem says what a price *means* once one exists; the next result says when one exists.

### 3.3 The price curve

Let  $c_0 = C_\mu(q^0)$  be the cost of the uniform rows, and let  $c_{\min} = \sum_i \mu_i \min_j u_{ij}$  be the smallest achievable cost.

**Proposition 4** (Price regimes). *Suppose some row with  $\mu_i > 0$  has nonconstant costs. Then  $\lambda \mapsto C_\mu(q^\lambda)$  is continuous on  $[0, \infty)$ , with*

$$\frac{d}{d\lambda} C_\mu(q^\lambda) = - \sum_i \mu_i \text{Var}_{q_{i\cdot}^\lambda}(u_{i\cdot}) < 0,$$

*and it decreases from  $c_0$  at  $\lambda = 0$  to  $c_{\min}$  as  $\lambda \rightarrow \infty$ . The budget problem has five regimes:*

(a)  $b < c_{\min}$ : *infeasible.*

(b)  $b = c_{\min}$ : *every optimizer is uniform on the cost minimizers of each row with  $\mu_i > 0$  (rows with  $\mu_i = 0$  are unconstrained); the Gibbs limit  $\lambda \rightarrow \infty$ , uniform on every row's minimizers, is one optimizer, and no finite price attains it.*

(c)  $c_{\min} < b < c_0$ : there is a unique finite price  $\lambda^*(b) > 0$  with  $C_\mu(q^{\lambda^*}) = b$ . On this interval  $V'(b) = \lambda^*(b)$  and  $\frac{d\lambda^*}{db} = -\left[\sum_i \mu_i \text{Var}_{q_i^{\lambda^*}}(u_i)\right]^{-1} < 0$ .

(d)  $b = c_0$ : the constraint is active and the price is zero.

(e)  $b > c_0$ : the constraint is slack, the price is zero, and the uniform rows are optimal.

If every row with  $\mu_i > 0$  has constant costs, then  $c_0 = c_{\min}$  and the price is not identifiable.

Two things follow. First, the anti-monotone price curve in Section 5.3 is guaranteed by theorem, so the experiment checks the solver rather than discovering a law. Second, “price emerges when the budget binds” needs one refinement: the price is positive exactly when the budget lies strictly inside  $(c_{\min}, c_0)$ . At the threshold  $b = c_0$  the budget binds with price zero. If one instead imposes an equality constraint  $C_\mu(q) = b$ , then for  $c_0 < b < c_{\max}$ , with  $c_{\max} = \sum_i \mu_i \max_j u_{ij}$ , the Gibbs multiplier is finite and negative; at  $b = c_{\max}$  it requires the limit  $\lambda \rightarrow -\infty$ , and for  $b > c_{\max}$  the equality is infeasible. Nonnegative prices belong to inequality budgets.

### 3.4 Canonical instances

The classical examples have this form. In equilibrium statistical mechanics, energy is the ledger, entropy is maximized under a mean-energy constraint, and the multiplier is the inverse temperature  $\beta$  (in units where entropy is dimensionless); free energy packages the trade-off [8, 14]. In rate–distortion theory, rate is minimized subject to a distortion budget; the nonnegative multiplier of that budget is, wherever the rate–distortion function is differentiable, the negative of its slope, and it trades fidelity against code length [2, 5]. This is an instance of general constrained duality rather than of the fixed-cost entropy programme above. In economics, the budget is the direct feasibility currency, and prices are the dual variables of resource and clearing constraints [1]. The same structure describes regularization strengths as prices of model complexity and, in rational-inattention models, a price on attention or information-processing capacity [18].

What these share is the geometry of constrained duality: a ledger, a budget, and a supporting multiplier. What they do not share is the source of the budget. In each case it comes from a domain-specific resource law, which is exactly the role of Assumption 2. The principle explains why prices follow once such a law is in place. It does not claim that every stable layer has one.

### 3.5 Relation to PICA

In the PICA interaction algebra the relevant cell is **P2**←**P6**: accounting informs gating [20]. That cell is the local form of the first half of the principle, in which a ledger becomes a feasibility boundary. Two neighbouring cells are suggestive: **P5**←**P6**, in which audit reshapes packaging (Section 5.6 gives a quantitative instance), and **P6**←**P6**, in which accounting regulates itself. PICA supplied a laboratory in which primitives can be switched on and off. Here we read that laboratory economically: what closures spend, how the spending becomes a constraint, and when the constraint returns a price.

## 4 Audits that observation cannot inflate

Audits measure irreversible drive, and they are natural candidates for the costs that enter a ledger. Whether a given audit quantity really is the maintenance spending of a layer is a modelling question that no inequality settles. What can be proved is that the audit is honest with respect to observation. This section fixes the two audits used in the laboratory, path-reversal asymmetry and cycle affinities, and proves that coarse observation can hide the first but never create it.

#### 4.1 Path-reversal asymmetry

Let  $P$  be a row-stochastic kernel on a finite set  $Z$  [9], let  $\rho_0$  be an initial law, and let  $\mathbb{P}_{\rho_0, T}$  be the law of the path  $x_{0:T} = (x_0, \dots, x_T)$ . Let  $\mathcal{R}$  reverse the order of a path. The directionality audit is

$$\Sigma_T = D_{\text{KL}}(\mathbb{P}_{\rho_0, T} \parallel \mathcal{R}_* \mathbb{P}_{\rho_0, T}) \in [0, \infty],$$

the standard forward-versus-reversed relative entropy of stochastic thermodynamics [11, 12, 16]. When every path of positive probability has a reversal of positive probability,

$$\Sigma_T = \mathbb{E}[\log \rho_0(X_0) - \log \rho_0(X_T)] + \sum_{t=0}^{T-1} \mathbb{E} \left[ \log \frac{P(X_t, X_{t+1})}{P(X_{t+1}, X_t)} \right].$$

The first term is a boundary term and cannot be dropped for a nonstationary initial law. A reversible kernel gives  $\Sigma_T = 0$  when it is started from its stationary law, but in general not from an arbitrary  $\rho_0$ . If some path of positive probability has a reversal of probability zero, for example because it uses a one-way edge, then  $\Sigma_T = \infty$ .

**Theorem 5** (Data processing for path audits). *Let  $\alpha, \beta$  be finite sets,  $F: \alpha \rightarrow \beta$  any map, and  $p, q$  probability laws on  $\alpha$  with  $\text{supp } p \subseteq \text{supp } q$ . Then*

$$D_{\text{KL}}(F_* p \parallel F_* q) \leq D_{\text{KL}}(p \parallel q).$$

*In particular, if  $p$  is a law on paths of length  $n$  with  $\text{supp } p \subseteq \text{supp } \mathcal{R}_* p$ , and  $f$  is applied pointwise to every state of the path, then*

$$D_{\text{KL}}(f_* p \parallel \mathcal{R}_* f_* p) \leq D_{\text{KL}}(p \parallel \mathcal{R}_* p).$$

The theorem is the classical data-processing inequality [5]. The path form uses only the fact that pointwise observation commutes with reversal. Neither surjectivity nor a full-support reference law is needed, and the reference may vanish wherever the first law does. Both statements are proved in Lean (Appendix B). Without the support hypothesis the right-hand side is  $+\infty$ , so the inequality holds trivially in the extended reals; that extended form is standard but is not part of the mechanization.

#### 4.2 Empirical audits and a regularization that keeps the inequality

From data, we estimate  $\Sigma_T$  on sliding windows of length  $T + 1$ . Let  $\hat{p}$  be the empirical law of the observed windows. If some observed window was never seen reversed, then  $D_{\text{KL}}(\hat{p} \parallel \mathcal{R}_* \hat{p}) = \infty$ . This is the honest answer. Discarding such windows and summing the rest does not give a KL divergence, and it need not obey data processing. With ten windows, nine of type  $(0, 1)$  and one of type  $(2, 3)$ , the truncated micro sum is zero, but observing the states through the labels  $(0, 1, 1, 0)$  yields a finite observed audit  $0.8 \log 9 > 0$ .

To compare finite numbers we regularize by mixing in the reversal.

**Proposition 6** (Reversal-mixture audit). *For  $0 < \varepsilon < \frac{1}{2}$  and any law  $p$  on windows, let  $p_\varepsilon = (1 - \varepsilon)p + \varepsilon \mathcal{R}_* p$  and*

$$\Sigma_{T, \varepsilon}(p) = D_{\text{KL}}(p_\varepsilon \parallel \mathcal{R}_* p_\varepsilon).$$

*Then  $\Sigma_{T, \varepsilon}(p)$  is finite,  $\Sigma_{T, \varepsilon}(p) \leq D_{\text{KL}}(p \parallel \mathcal{R}_* p)$ , and for every pointwise observation  $f$ ,*

$$\Sigma_{T, \varepsilon}(f_* p) \leq \Sigma_{T, \varepsilon}(p).$$

Observation commutes with the mixture, so the inequality holds for the *same* sample and needs no statistical argument (Appendix A). The regularized audit is a lower bound on the raw audit. It is not an unbiased estimator of the population  $\Sigma_T$ , and we do not use it as one. We report the raw missing-reverse mass, the probability under  $\hat{p}$  of windows whose reversal was never observed, next to it.

### 4.3 Cycle affinities

The second audit lives on loops. On each edge whose two directions both have positive probability, define the antisymmetric form  $a_{ij} = \log(P_{ij}/P_{ji})$ . For an oriented cycle  $\gamma$  in the undirected support graph of these bidirectional edges, the affinity is

$$\mathcal{A}(\gamma) = \sum_{(i \rightarrow j) \in \gamma} a_{ij}$$

[7, 15]. For a simple graph without self-loops, the cycle space has dimension  $\beta_1 = m - n + c$  (edges minus vertices plus components). The map from edge forms to cycle integrals has rank  $\beta_1$ , and all cycle integrals vanish exactly when  $a$  is the gradient of a vertex potential on each component. For bidirectional kernels this is Kolmogorov’s criterion for reversibility. One-way edges are excluded from this audit and are left to the path audit, which registers them as infinite asymmetry whenever a path of positive probability within the audited horizon traverses one. The affinities of a cycle basis are low-dimensional drive coordinates. Their Euclidean norm depends on the chosen basis, so we use norms only to compare drive within one fixed basis.

Two dimensions should not be confused. The cycle rank  $\beta_1$  counts the loop directions *available* at a resolution. The rank of the map from a model’s drive parameters to cycle affinities counts the directions *actually driven*. Section 5.4 shows that the two can differ.

## 5 A finite laboratory

We now illustrate each part of the chain in Figure 1 on finite Markov chains. The point is not to discover the theorems of Sections 3 and 4 empirically, since they need no such support. It is to show them operating on concrete substrates, to check the numerical pipeline against them, and to isolate the effects that depend on the chosen example rather than on a theorem.

### 5.1 Substrates, lenses, and protocol

**Module-ring chain.** The main substrate has  $M = 6$  rings of  $L = 10$  states, 60 states in all. On ring  $m$  each state sends weight  $e^{+A_m/(2L)}$  clockwise and  $e^{-A_m/(2L)}$  counter-clockwise, and neighbouring rings are joined at their first state by a symmetric bridge of weight 0.2; rows are then normalized. Row normalization adds only a gradient to the edge log-ratios, so each ring carries cycle affinity exactly  $A_m$ . We use  $A_m$  evenly spaced from 1.0 to  $-0.5$ . Every edge is bidirectional, so all path audits of the true chain are finite. Trajectories start from the uniform law.

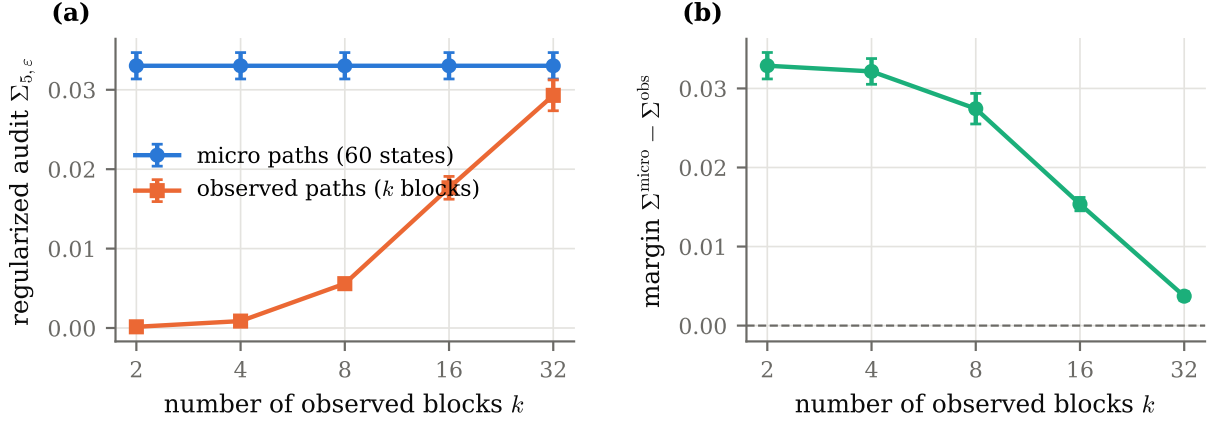
**Ladder chain.** For the resolution ladder we use the same rings joined by *two* bridges per neighbouring pair, at positions 0 and  $L/2$ , again with weight 0.2.

**Lenses.** A lens is a partition  $f : Z \rightarrow X$  of the states into blocks. To audit observed trajectories we apply  $f$  pointwise to each sampled path. When a Markov kernel at the coarse level is needed, as for the ladder, we use stationary-conditional lumping [4, 9],

$$Q_{ab} = \sum_{i:f(i)=a} \pi(i | a) \sum_{j:f(j)=b} P_{ij},$$

where  $\pi$  is the stationary law of  $P$ . The lumped kernel summarizes the coarse support structure. It is not used for any data-processing claim, because the lumped chain is generally not the law of the observed paths.

**Protocol.** Every design choice (substrate, partition seed, proxy permutation, sample sizes, regularization) is fixed before running, and every run is reported. The five “seeds” mean different things in different experiments. For the path audit and the proxy comparison they are



**Figure 2:** Coarse observation hides directionality but never creates it. (a) Reversal-mixture path audit  $\Sigma_{5,\varepsilon}$  ( $\varepsilon = 0.01$ ) of micro trajectories and of the same trajectories observed through  $k$  contiguous blocks. (b) Margin between the two. Points are means over five independent samples; bars are standard deviations. The margin is nonnegative on every sample by Proposition 6.

independent samples from one fixed design. For the price curve each seed also draws its own random partition and estimates its own cost. The ladder computation is deterministic, so its seed labels are repetitions. All numbers below come from the receipts listed in Appendix C.

## 5.2 Coarse observation never inflates the audit

We sample  $N = 2000$  trajectories of length 80 from the module-ring chain and compute the reversal-mixture audit  $\Sigma_{5,\varepsilon}$  (windows of six states,  $\varepsilon = 0.01$ ) on the micro paths and on the same paths observed through  $k \in \{2, 4, 8, 16, 32\}$  contiguous blocks.

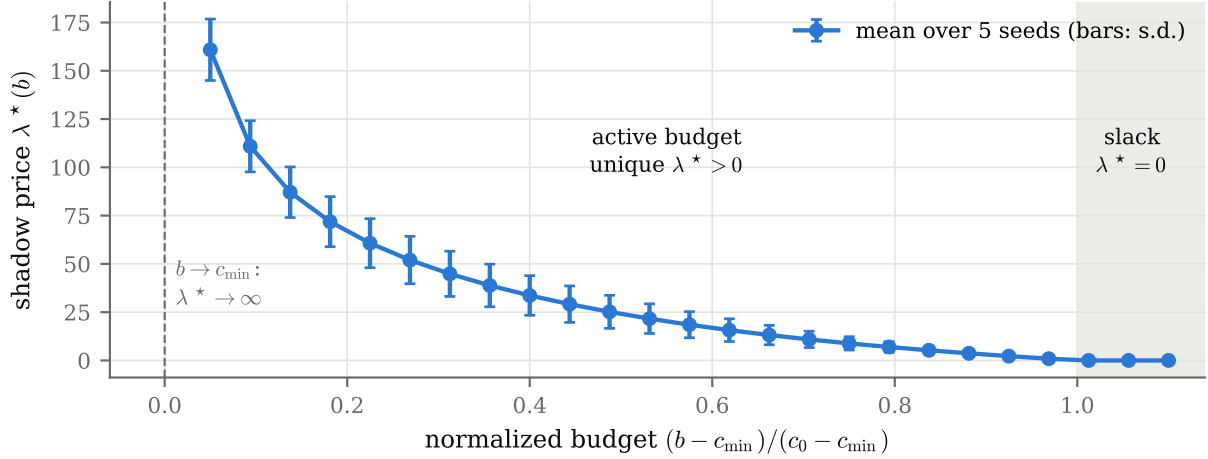
The micro audit is  $0.0330 \pm 0.0017$  (mean  $\pm$  s.d. over five seeds). The observed audit rises with resolution, from  $1.5 \times 10^{-4}$  at  $k = 2$  to 0.0293 at  $k = 32$ , and stays below the micro audit in every seed, with the smallest mean margin 0.00372 at  $k = 32$  (Figure 2). Proposition 6 guarantees this on every sample, so the result is a consistency check of the implementation, not an independent discovery. It also shows the other half of the story: coarse observation can hide most of the drive, since at  $k = 2$  almost none is visible.

The raw micro empirical audits are infinite in every seed: some micro windows were never seen reversed, with mean missing-reverse mass 0.00205. Raw observed audits are infinite whenever their missing-reverse mass is positive; its mean over seeds falls from 0.00145 at  $k = 32$  to  $2.3 \times 10^{-5}$  at  $k = 2$ . These are sampling gaps, not one-way edges, since the true chain is fully bidirectional. They are exactly why the regularized audit is needed for a finite comparison.

## 5.3 The price of an inherited budget

**Cost.** We partition the module-ring chain into  $k = 8$  random blocks, sample 8000 trajectories of length 80, and give each micro transition the cost  $\max\{0, \log(P_{ij}/P_{ji})\}$ . The macro cost  $u_{ab}$  is the average of this cost over observed transitions from block  $a$  to block  $b$ , shrunk toward the global mean by one pseudo-observation so that unobserved macro transitions have a defined cost.

This ledger is a modelling choice, and it should not be confused with entropy production. The log-ratio of a single edge depends on how the rows are normalized: it changes by a gradient  $\phi_j - \phi_i$  under a change of state potentials, and only its sums around cycles are invariant. The stationary entropy production instead uses  $\log[\pi_i P_{ij}/(\pi_j P_{ji})]$ . Moreover, the macro Gibbs array has full support and may place mass on block transitions that the micro chain never makes. The experiment is therefore a synthetic closure on an estimated cost array; it does not construct micro maintenance protocols for each macro transition.



**Figure 3:** Shadow price of an inherited budget. The horizontal axis is the budget normalized so that 0 is the cheapest feasible cost  $c_{\min}$  and 1 is the cost  $c_0$  of the unconstrained (uniform) closure. Points are means over five seeds, each with its own random partition and estimated cost; bars are standard deviations. Shading marks the slack regime, where the price is exactly zero.

**Sweep.** With uniform row weights, we solve the budget problem of Theorem 3 for 25 budgets  $b = c_{\min} + \alpha(c_0 - c_{\min})$ , with  $\alpha$  evenly spaced from 0.05 to 1.10. Figure 3 shows the result. The price falls from  $160.9 \pm 15.9$  at  $\alpha = 0.05$  to  $0.90 \pm 0.35$  at  $\alpha = 0.969$ , and is exactly zero for the three budgets with  $\alpha > 1$  in all five seeds. In the active region the solver meets the budget to within  $3 \times 10^{-13}$  in every seed. In the slack region the achieved cost stays at  $c_0$ , below the budget. The per-seed Spearman correlation between budget and price is  $-0.99923$ ; it differs from  $-1$  only because the three slack prices are tied at zero.

All of this is predicted by Proposition 4: a strictly decreasing price on  $(c_{\min}, c_0)$ , divergence as  $b \rightarrow c_{\min}$ , and zero price above  $c_0$ . The experiment adds no evidence for the shape of the curve. What it shows is that a cost read off a lower-layer ledger gives a nondegenerate price at the next layer. The cost is nonconstant within rows, so the price is identifiable, and the curve spans two orders of magnitude before it vanishes.

#### 5.4 Resolution adds loop directions, not necessarily drive

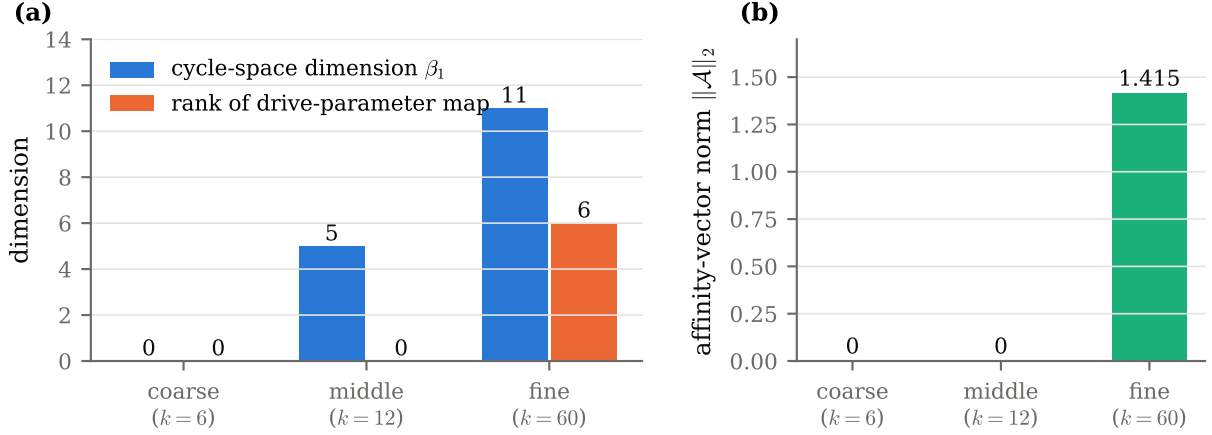
On the ladder chain we compare three lenses: whole rings ( $k = 6$ ), half-rings ( $k = 12$ ), and single states ( $k = 60$ ). For each, we form the lumped kernel, its bidirectional support graph, a cycle basis, and the cycle affinities.

**Proposition 7** (Ladder cycle ranks). *For  $M$  rings of even length  $L \geq 4$  with positive bridges at positions 0 and  $L/2$ , the three support graphs have cycle ranks 0,  $M - 1$ , and  $2M - 1$ .*

For  $M = 6$  this gives 0, 5, 11, which is what the code reports. The construction is deterministic, so the five seed labels repeat a single computation rather than test robustness. The proof is a count of edges (Appendix A).

These are the dimensions of the *available* cycle coordinates. The dimension actually *driven* is smaller (Figure 4). The model has only  $M = 6$  drive parameters  $A_1, \dots, A_6$ . At the fine level, each cycle affinity is a linear function of these parameters, with coefficients given by signed traversal counts divided by  $L$ . Whole-ring cycles recover each  $A_m$ , so the map from parameters to affinities has rank 6, not 11. At the half-ring level the cycle space has dimension 5, but every affinity vanishes to rounding error (norm  $6.7 \times 10^{-15}$ ) and the parameter map has rank 0. Translating every ring by half a turn swaps the two halves of each ring and preserves the kernel and its stationary law, and under this symmetry the affinities of the rectangular cycles cancel. Only the fine lens sees nonzero drive (affinity norm 1.415 in the computed basis).

The ladder therefore separates capacity from activity. Finer resolution exposed more loop directions in which a drive currency *could* live, but realized drive appeared only where the



**Figure 4:** Available versus active drive coordinates on the ladder chain. (a) Cycle-space dimension  $\beta_1$  of the lumped support graph and rank of the linear map from the six ring affinities  $A_1, \dots, A_6$  to cycle affinities. (b) Euclidean norm of the cycle-affinity vector in the computed basis, which is basis-dependent and is shown for comparison within each level only. The half-ring level has five independent loops but no drive.

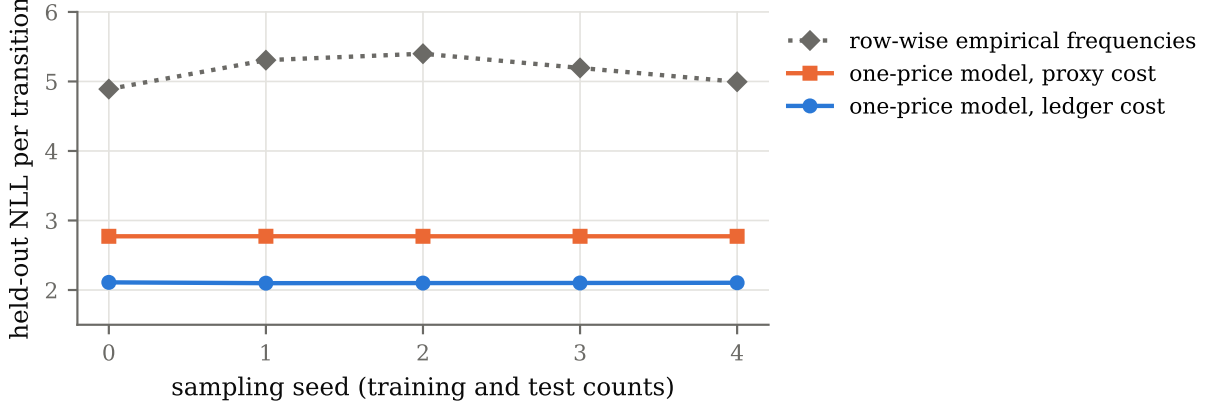
substrate actually carries it. Nor does finer resolution always add loops. Take a reversible chain on the four-vertex path  $0 - 1 - 2 - 3$ , a tree with  $\beta_1 = 0$ , and lump the endpoints 0 and 3 into one block. The lumped support graph is a triangle with  $\beta_1 = 1$ . The growth  $0 \rightarrow 5 \rightarrow 11$  is a property of this construction, not a law about refinement.

### 5.5 A ledger cost predicts better than a proxy

Is any low-dimensional penalty as good as a ledger-derived one? We build a ledger cost  $u^{\text{led}}$  as in Section 5.3, now with  $k = 16$  random blocks, and take as ground truth the Gibbs kernel  $q^{\text{true}}$  at the budget  $c_{\min} + 0.35(c_0 - c_{\min})$ . The proxy  $u^{\text{prx}}$  rearranges the entries of each row of  $u^{\text{led}}$  by a row-specific random permutation; the permutations are drawn once from a fixed seed and used for every sample. It therefore has exactly the same values in every row, attached to the wrong transitions.

For each of five samples we draw 30 training and 5000 test transitions per row and compare three models: row-wise empirical frequencies (clipped below  $10^{-12}$  and renormalized), and one-price Gibbs models fitted with  $u^{\text{led}}$  and with  $u^{\text{prx}}$ . The price is fitted by maximizing the conditional-logit likelihood  $\ell(\lambda) = \sum_{ij} C_{ij} \log q_{ij}^\lambda$  [13]. This likelihood is concave, with curvature  $\sum_i N_i \text{Var}_{q_i^\lambda}(u_i)$ , where  $N_i$  is the number of training transitions from row  $i$ . The price is identifiable when some observed row has nonconstant cost. If some observed row has nonconstant cost but every observed transition already minimizes its row's cost, the likelihood keeps increasing and the fitted price is  $+\infty$ . An optimum at the boundary  $\lambda = 0$  can occur, and then the inverse curvature is not a symmetric confidence interval.

**Prediction.** The ledger model has the lower held-out negative log-likelihood in every sample, with a mean of 2.103 per transition against 2.773 for the proxy and 5.157 for empirical frequencies (Figure 5). The empirical baseline does badly because 30 draws over 16 destinations leave many transitions unseen, which then receive probability near zero. The proxy model does no better than uniform guessing, whose loss is  $\log 16 \approx 2.7726$ . The direction of the gap is expected. Since the data come from  $q^{\text{true}}$ , any fixed kernel  $r$  has expected excess held-out loss per transition  $\sum_i w_i D_{\text{KL}}(q_i^{\text{true}} \| r_i) \geq 0$  over the truth, where  $w_i$  is the share of test transitions from row  $i$ . This excess is strictly positive unless  $r$  agrees with the truth on every row with  $w_i > 0$ . A model family that contains the truth, as the ledger family does, can drive that excess toward zero as the training data grow. The proxy family contains the truth only if its permuted costs happen to induce the same Gibbs rows, which they do not here.



**Figure 5:** Held-out negative log-likelihood per transition for five independent samples from a fixed design. The proxy cost has exactly the same values in each row as the ledger cost but attaches them to the wrong transitions; its fit is close to the uniform model, whose loss is  $\log 16 \approx 2.7726$ .

**Prices.** The fitted ledger prices range from 62.1 to 70.3 (s.d. 3.18), while the proxy prices range from 0 to 1.67 (s.d. 0.55), with one sample at the boundary  $\lambda = 0$ . The proxy’s smaller spread is not a sign of a better currency. Its fits stay near the uniform model, and prices attached to different cost coordinates do not share a scale. Prediction, not across-sample price variability, is the comparison this design supports.

### 5.6 Pricing exits buys object stability

Finally we connect the price to objecthood itself. Partition the states into blocks with  $f$ , and let  $\pi$  be a stationary law of  $P$  with positive mass  $\pi(a)$  on every block  $a$ . Packaging pushes a distribution forward to blocks and lifts it back with the stationary prototypes  $\nu_a = \pi(\cdot | a)$ . For an integer number of steps  $\tau \geq 0$ ,

$$\mathcal{B}(\mu) = \sum_a \mu(f^{-1}(a)) \nu_a, \quad E_\tau(\mu) = \mathcal{B}(\mu P^\tau), \quad \delta(\mu) = \|E_\tau(E_\tau(\mu)) - E_\tau(\mu)\|_1.$$

The map  $\mathcal{B}$  is stochastic and idempotent. The defect  $\delta \in [0, 2]$  measures how much a packaged object moves when it is evolved and repackaged once more [21].

**Theorem 8** (Packaging defect bound). *Let  $\varepsilon = \max_i \sum_{j:f(j) \neq f(i)} P_{ij}$  be the largest one-step exit probability from a block. Then for every probability law  $\mu$ ,*

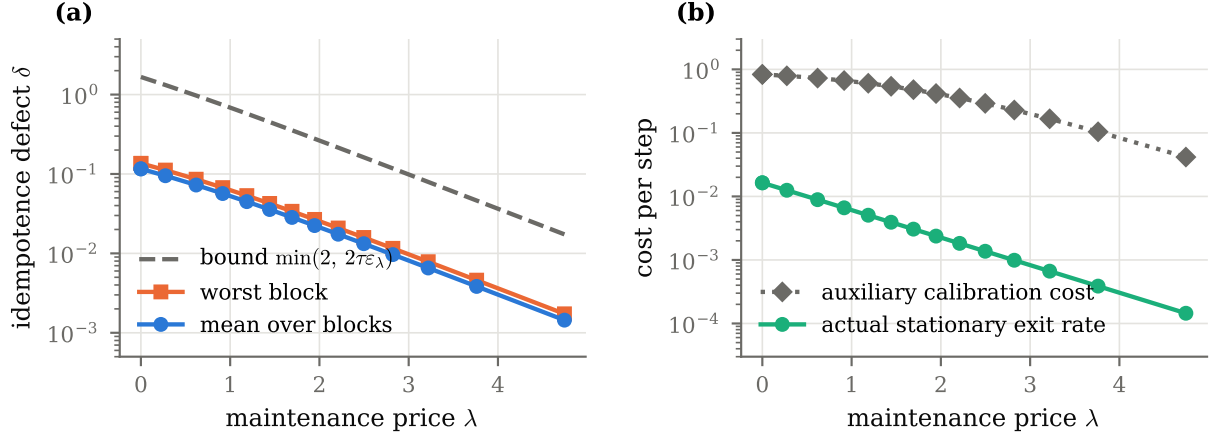
$$\delta(\mu) \leq \min(2, 2\tau\varepsilon).$$

The proof (Appendix A) uses stationarity: the mass a prototype loses from its block in one step equals the mass flowing into the block from outside, so one step moves  $\nu_a$  by at most  $2\varepsilon$  in  $\ell_1$  distance. This adds up to at most  $2\tau\varepsilon$  over  $\tau$  steps, and the bound extends to all lifted laws by convexity. The argument needs the stationary prototype; it is not claimed for uniform prototypes of a nonuniform block.

Now price the exits. Multiplying every inter-block transition by  $e^{-\lambda}$  and renormalizing gives  $P^{(\lambda)}$ . If row  $i$  keeps mass  $r_i$  inside its block and sends  $s_i = 1 - r_i$  out, its exit probability becomes

$$\varepsilon_i(\lambda) = \frac{s_i e^{-\lambda}}{r_i + s_i e^{-\lambda}} \leq \frac{s_i}{r_i} e^{-\lambda}.$$

When every  $r_i > 0$ , as in the module-ring chain, Theorem 8 gives  $\delta \leq 2\tau \max_i (s_i/r_i) e^{-\lambda} \rightarrow 0$ . For a large enough exit price, every packaged object is therefore uniformly close to idempotent. The bound does not imply that the defect decreases monotonically in  $\lambda$  for every kernel family, because the prototypes change with  $\lambda$ . If some row has  $r_i = 0$ , no vanishing-exit conclusion follows.



**Figure 6:** Pricing block exits makes packaged objects nearly idempotent. (a) Idempotence defect of the stationary block prototypes (mean and worst block) and the bound  $\min(2, 2\tau\epsilon_\lambda)$  of Theorem 8, with  $\tau = 5$ . (b) The auxiliary calibration cost used to choose  $\lambda$ , compared with the actual stationary exit rate of the controlled chain. Prices are calibrated, not inferred from spending.

**Experiment.** We take the module-ring chain with rings as blocks ( $k = 6$ ) and  $\tau = 5$ . The prices are calibrated with an auxiliary  $6 \times 6$  Gibbs problem whose cost is 1 for leaving a block and 0 for staying, with uniform row weights. Fifteen budgets between 0.042 and 0.917 give prices from 4.74 down to 0. The two budgets above  $c_0 = 5/6$  are slack. We report the defect at each block prototype; by linearity of  $E_\tau$ , the largest of these is the worst defect over all lifted laws.

The mean defect over blocks falls from 0.116 at  $\lambda = 0$  to 0.00144 at  $\lambda = 4.74$ , and the worst block falls from 0.136 to 0.00173 (Figure 6a). Every value lies below the bound  $2\tau\epsilon_\lambda$ , which falls from 1.67 to 0.0174, and the defect decreases along the whole sampled sweep.

The calibration budget should not be read as the substrate’s spending. The auxiliary problem lives on a complete  $6 \times 6$  block graph, whereas the controlled chain can leave a ring only through its bridges. Its actual stationary exit rate,  $\sum_i \pi_\lambda(i) \sum_{j: f(j) \neq f(i)} P_{ij}^{(\lambda)}$ , runs from 0.0164 down to  $1.4 \times 10^{-4}$ , between about 50 and 300 times below the calibration cost (Figure 6b). The experiment thus demonstrates enforcement by a calibrated price, with actual spending measured separately.

## 6 Discussion

### 6.1 Why inverse temperatures, prices, and regularizers are the same kind of object

Inverse temperature, market price, regularization strength, and attention-like precision are different things. In form, however, each is a multiplier that prices a limited resource in a layer that closes by constrained optimization [1, 8, 18]. Theorem 3 makes the shared form exact in the finite entropy setting. Whatever the substrate, once a ledger is budgeted and the layer selects the least-committed law within the budget, the Gibbs multiplier is a supporting price of that budget. When some weighted row has nonconstant costs, it is positive exactly when the budget lies strictly between the cheapest feasible cost and the cost of the unconstrained closure (Proposition 4).

This is more specific than saying that Lagrange multipliers exist. The usual statement starts from a given programme. The currency–constraint principle states where the programme’s constraint comes from: a maintenance ledger together with a resource law (Assumption 2). It also states what happens to the price as the law tightens or relaxes. The principle is directional. Lower-layer spending becomes a higher-layer constraint, and the constraint becomes a higher-layer price. Layers are linked by maintenance, not merely by a formal duality.

## 6.2 What the principle does not say

The principle does not derive budgets from persistence. Example 1 shows that a perfectly persistent object can cost nothing, and an external ledger attached to it can be unbounded. A resource law is a separate physical, economic, or computational fact about the substrate. Conservation, a finite supply, a viability requirement, or a sustainability constraint can each provide one. Where such a law exists and the layer closes by maximum entropy with fixed row weights, a feasible budget has a price determined by Theorem 3 and Proposition 4. If some weighted row has nonconstant costs, the price is finite and nonnegative above the cheapest feasible cost, and no finite price attains that cost itself; otherwise the price is not identifiable. Where no resource law exists, the principle is silent.

Nor does the principle say that every layer closes by maximum entropy, or that one scalar price suffices. The laboratory uses a single-constraint Gibbs family because it is the simplest closure in which the price can be computed and tested. Several budgets would give a vector of prices, and other closures would give other duals.

## 6.3 Capacity, drive, and resolution

The ladder experiment makes a distinction that is easy to blur. A finer lens can expose more independent loops, and hence more coordinates in which a drive currency *could* be expressed, without any additional drive being present. In the ladder construction the half-ring lens has five independent loops and zero affinity, because of a symmetry of the construction. The fine lens has eleven loops but only six driven directions, because the model has only six drive parameters. Coarsening can even create loops, as the path-to-triangle example shows. Statements about the “dimension of the currency space” at a given resolution should therefore say whether they mean available cycle coordinates or the rank of the actual drive.

## 6.4 What counts as evidence here

The laboratory is synthetic, and its role differs from claim to claim. For data processing, the price curve, and the packaging bound, the theorems carry the weight, and the numbers check that the implementation agrees with them. For the ladder, the ranks are exact properties of a specific construction, and the counterexample marks their limits. For the proxy comparison, the population statement (a fixed wrong kernel has positive expected excess loss) is a theorem, and the experiment shows a large gap at a modest sample size. The variability of fitted prices across samples is not a meaningful comparison between different cost coordinates, so it is not used as evidence for either cost.

Several modelling choices limit interpretation. The macro ledger uses the positive part of single-edge log-ratios, which is not invariant under a change of state potentials and is not the stationary entropy production. The macro Gibbs closures have complete support and need not correspond to transitions the substrate can realize. In the packaging experiment the price is chosen by an auxiliary calibration rather than inferred from the substrate’s spending, and the substrate’s actual spending is about 50 to 300 times lower. Uncertainty statements from the conditional-logit curvature are curvature scales, not sampling guarantees, under misspecification or at the boundary  $\lambda = 0$ .

## 6.5 What this paper adds to the Six Birds series

The foundations paper set out the vocabulary of closure: lenses, packaging, definability, and audits [21]. PICA supplied an interaction algebra and a laboratory in which primitives can be enabled and coupled [20]. This paper adds the spending side. It identifies what a closure must budget to keep its objects, gives the exact conditions under which that budget becomes a constraint and the constraint a price, and supplies audits that coarse observation cannot fake. The Lean development makes the audit inequality and the single-row duality machine-checked, and the packaging bound connects the price of leaving a block to how stable the block is as an object.

## 7 Conclusion

Mature theories keep introducing small sets of variables that bound feasibility, add up along protocols, or act as exchange rates between competing aims. Within the Six Birds closure calculus we have treated these variables as currencies, defined by the structural role they play relative to a layer.

The central result is conditional and exact. Suppose a lower-layer ledger must stay within a budget for higher-layer objects to be maintained. Then the ledger is a feasibility constraint for the higher layer. If that layer closes by maximum entropy, the constraint's Gibbs multiplier is a nonnegative shadow price. Provided some weighted row has nonconstant costs, that price is positive exactly when the budget lies strictly between the cheapest feasible cost and the unconstrained cost, zero at or above the unconstrained cost, and divergent as the budget approaches the cheapest feasible cost; with constant costs the price is not identifiable. Path-reversal KL, and its reversal-mixture regularization on finite samples, cannot be inflated by coarse observation. And with stationary prototypes, a fixed horizon, and positive within-block mass in every row, the uniform bound on packaging defects tends to zero as the price of leaving a block grows.

A finite Markov laboratory showed each step at work. It also marked where the evidence is a theorem, where it is a property of one construction, and where an apparent pattern does not hold in general. The broader picture follows. Wherever stable description must be maintained on a limited resource, budgets appear, and where budgets bind in a maximum-entropy closure, prices appear. Persistence alone does not supply the budget; given a feasible budget and such a closure, the price follows.

## A Proofs

Throughout,  $0 \log 0 = 0$ , and  $D_{\text{KL}}(p \| q) = \sum_x p(x) \log(p(x)/q(x))$  whenever  $\text{supp } p \subseteq \text{supp } q$  (and  $+\infty$  otherwise).

### A.1 Theorem 3 (finite Gibbs duality)

(i) Since  $\log q_{ij}^\lambda = -\lambda u_{ij} - \log Z_i(\lambda)$  and  $q^\lambda$  is strictly positive, for each row

$$D_{\text{KL}}(p_i \| q_i^\lambda) = \sum_j p_{ij} \log p_{ij} + \lambda \sum_j p_{ij} u_{ij} + \log Z_i(\lambda),$$

using  $\sum_j p_{ij} = 1$ . Multiplying by  $\mu_i$  and summing gives (i).

(ii) Setting  $p = q^\lambda$  in (i) gives  $0 = -H_\mu(q^\lambda) + \lambda C_\mu(q^\lambda) + \sum_i \mu_i \log Z_i$ . Subtracting this from (i) for a general  $p$ , and using that the left side of (i) is nonnegative, gives (ii).

(iii) If  $C_\mu(p) \leq b$  and  $\lambda \geq 0$ , then by (ii) and complementary slackness,

$$H_\mu(p) \leq H_\mu(q^\lambda) + \lambda(C_\mu(p) - C_\mu(q^\lambda)) \leq H_\mu(q^\lambda) + \lambda(b - C_\mu(q^\lambda)) = H_\mu(q^\lambda).$$

Since  $q^\lambda$  is feasible, it is optimal.

(iv) The same chain with  $C_\mu(p) \leq b'$  gives  $H_\mu(p) \leq H_\mu(q^\lambda) + \lambda(b' - b)$ . By (iii),  $V(b) = H_\mu(q^\lambda)$ , so taking the supremum over  $p$  gives  $V(b') \leq V(b) + \lambda(b' - b)$ . The function  $V$  is concave, because the objective is concave and the constraint is linear in  $(p, b)$ . A concave function differentiable at  $b$  has its derivative as its only supergradient.

*Uniqueness.*  $H_\mu$  is strictly concave in the rows with  $\mu_i > 0$ , and the feasible set is convex. If two optimizers differed on such a row, their midpoint would be feasible with strictly larger entropy.  $\square$

## A.2 Proposition 4 (price regimes)

Write  $C(\lambda) = C_\mu(q^\lambda) = \sum_i \mu_i \mathbb{E}_{q_i^\lambda}[u_i]$ . Differentiating the Gibbs mean gives  $\frac{d}{d\lambda} \mathbb{E}_{q_i^\lambda}[u_i] = -\text{Var}_{q_i^\lambda}(u_i)$ . Because  $q^\lambda$  has full support, this variance is positive on every nonconstant row. As  $\lambda \rightarrow \infty$ ,  $q_i^\lambda$  converges to the uniform law on  $\arg \min_j u_{ij}$ , so  $C(\lambda) \rightarrow c_{\min}$ ; and  $C(0) = c_0$ .

(a) Every  $p$  has  $C_\mu(p) \geq c_{\min}$ . (b) At  $b = c_{\min}$  every feasible  $p$  puts all the mass of each positive-weight row on that row's minimizers, and entropy is maximized by the uniform law on them; zero-weight rows are unconstrained. The limit of  $q^\lambda$  is one such optimizer. It is not attained at finite  $\lambda$ , because  $q^\lambda$  charges every destination and some positive-weight row has non-minimizers. (c) By continuity and strict monotonicity,  $C(\lambda) = b$  has a unique solution  $\lambda^*(b) \in (0, \infty)$ , and  $q^{\lambda^*}$  is optimal by Theorem 3(iii). From (i) with  $p = q^\lambda$ ,  $H_\mu(q^\lambda) = \lambda C(\lambda) + \sum_i \mu_i \log Z_i(\lambda)$ . Differentiating, and using  $\frac{d}{d\lambda} \log Z_i = -\mathbb{E}_{q_i^\lambda}[u_i]$ , gives  $\frac{d}{d\lambda} H_\mu(q^\lambda) = \lambda C'(\lambda)$ . Hence  $V'(b) = \lambda^*(b)$ , and by the inverse function theorem  $d\lambda^*/db = 1/C'(\lambda^*)$ . (d,e) For  $b \geq c_0$ ,  $\lambda = 0$  satisfies feasibility and complementary slackness, so the uniform rows are optimal; the constraint is active at  $b = c_0$  and strictly slack for  $b > c_0$ . Finally, if every positive-weight row is constant,  $C(\lambda)$  is constant,  $c_0 = c_{\min}$ , and no budget determines  $\lambda$ .  $\square$

## A.3 Theorem 5 (data processing)

Group the atoms of  $\alpha$  into the fibres  $F^{-1}(y)$ . On a fibre with  $Q_y = \sum_{x \in F^{-1}(y)} q(x) > 0$ , convexity of  $\varphi(t) = t \log t + 1 - t$  (Jensen's inequality with weights  $q(x)/Q_y$ ) gives

$$Q_y \varphi\left(\frac{P_y}{Q_y}\right) \leq \sum_{x \in F^{-1}(y)} q(x) \varphi\left(\frac{p(x)}{q(x)}\right),$$

where  $P_y = \sum_{x \in F^{-1}(y)} p(x)$ . Atoms with  $q(x) = 0$  have  $p(x) = 0$  by the support hypothesis and contribute nothing. Fibres with  $Q_y = 0$  contribute zero on the left. Summing over  $y$ , and using that  $\sum_x q(x) \varphi(p(x)/q(x)) = D_{\text{KL}}(p \| q)$  under the support hypothesis, gives the first claim. For paths, pointwise observation commutes with reversal,  $f \circ \mathcal{R} = \mathcal{R} \circ f$ , so  $f_* \mathcal{R}_* p = \mathcal{R}_* f_* p$ , and the first claim applies with  $q = \mathcal{R}_* p$ . This is the argument mechanized in Lean (Appendix B).  $\square$

## A.4 Proposition 6 (reversal-mixture audit)

$\mathcal{R}$  is an involution. Since  $\varepsilon$  and  $1 - \varepsilon$  are positive,  $\mathcal{R}_* p_\varepsilon = (1 - \varepsilon) \mathcal{R}_* p + \varepsilon p$  has support  $\text{supp } p \cup \text{supp } \mathcal{R}_* p$ , which contains  $\text{supp } p_\varepsilon$ . Hence  $\Sigma_{T, \varepsilon}(p) < \infty$ .

By joint convexity of KL,

$$\Sigma_{T, \varepsilon}(p) \leq (1 - \varepsilon) D_{\text{KL}}(p \| \mathcal{R}_* p) + \varepsilon D_{\text{KL}}(\mathcal{R}_* p \| p) = D_{\text{KL}}(p \| \mathcal{R}_* p),$$

because applying the bijection  $\mathcal{R}$  to both arguments does not change KL.

Finally, let  $F$  apply  $f$  to every state of a window. Then  $F \circ \mathcal{R} = \mathcal{R} \circ F$ , so

$$F_*(p_\varepsilon) = (1 - \varepsilon) F_* p + \varepsilon \mathcal{R}_* F_* p = (F_* p)_\varepsilon \quad \text{and} \quad F_*(\mathcal{R}_* p_\varepsilon) = \mathcal{R}_*(F_* p)_\varepsilon.$$

Theorem 5, applied to  $p_\varepsilon$  and  $\mathcal{R}_* p_\varepsilon$ , gives  $\Sigma_{T, \varepsilon}(F_* p) \leq \Sigma_{T, \varepsilon}(p)$ . When  $p$  is an empirical window law, the observed-window law is exactly  $F_* p$ , so the inequality holds for the same sample.  $\square$

## A.5 Theorem 8 (packaging defect)

Fix a block  $a$  and its prototype  $\nu_a = \pi(\cdot | a)$ . Write  $\Phi_a = \sum_{i \in a, j \notin a} \pi(i) P_{ij}$  for the stationary flow out of  $a$ . For  $j \in a$ , stationarity gives  $(\nu_a P)(j) - \nu_a(j) = -\pi(a)^{-1} \sum_{i \notin a} \pi(i) P_{ij} \leq 0$ . For  $j \notin a$ ,  $(\nu_a P)(j) = \pi(a)^{-1} \sum_{i \in a} \pi(i) P_{ij}$ . The total flow into  $a$  equals  $\Phi_a$  by stationarity, so

$$\|\nu_a P - \nu_a\|_1 = \frac{2\Phi_a}{\pi(a)} = 2 \sum_{i \in a} \nu_a(i) \sum_{j \notin a} P_{ij} \leq 2\varepsilon.$$

Since  $P$  contracts  $\ell_1$  distances between probability laws, telescoping gives  $\|\nu_a P^\tau - \nu_a\|_1 \leq 2\tau\varepsilon$ . Because  $\mathcal{B}\nu_a = \nu_a$  and  $\mathcal{B}$  also contracts  $\ell_1$ ,

$$\|E_\tau(\nu_a) - \nu_a\|_1 = \|\mathcal{B}(\nu_a P^\tau) - \mathcal{B}(\nu_a)\|_1 \leq 2\tau\varepsilon.$$

For any probability  $\mu$ ,  $E_\tau(\mu) = \sum_a w_a \nu_a$  is a lifted law, and  $E_\tau$  is linear. Hence

$$\delta(\mu) = \left\| \sum_a w_a (E_\tau(\nu_a) - \nu_a) \right\|_1 \leq 2\tau\varepsilon,$$

and  $\delta \leq 2$  holds trivially. The same linearity shows that, over lifted inputs  $\mu = \sum_a w_a \nu_a$ ,  $\delta(\mu) \leq \sum_a w_a \delta(\nu_a) \leq \max_a \delta(\nu_a)$ , with equality at a prototype. The exit formula in Section 5.6 follows from renormalizing the row  $(r_i, s_i e^{-\lambda})$ .  $\square$

## A.6 Proposition 7 (ladder cycle ranks)

The ladder chain is irreducible, so its stationary law is positive. Hence  $Q_{ab} > 0$  exactly when some edge of  $P$  runs from block  $a$  to block  $b$ ; the support graphs are simple and we omit self-loops. *Rings*: the  $M$  ring-blocks are joined only through bridges between neighbouring rings, giving a path with  $M$  vertices and  $M - 1$  edges, so  $\beta_1 = 0$ . *Half-rings*: each ring contributes two blocks joined by one edge (ring edges cross between the halves). The bridge at position 0 joins the first halves of neighbouring rings, and the bridge at  $L/2$  joins their second halves. This gives  $2M$  vertices and  $M + 2(M - 1) = 3M - 2$  edges in one component, so  $\beta_1 = M - 1$ . *States*:  $ML$  vertices,  $ML$  ring edges (each ring is a simple cycle since  $L \geq 4$ ), and  $2(M - 1)$  bridges, so  $\beta_1 = 2M - 1$ .  $\square$

## B Lean formalization

The formal development lives under `lean/` and is written in Lean 4 with mathlib [6, 19]. It uses toolchain v4.28.0 and the mathlib revision pinned in `lean/lake-manifest.json` (8f9d9cff). It builds with `lake build`. The file `lean/CheckAxioms.lean` prints the transitive axioms of every principal theorem, and all of them depend only on `propext`, `Classical.choice`, and `Quot.sound`. The development contains no `sorry`, `admit`, or added axioms.

### B.1 Finite KL and data processing (FiniteKL.lean)

The development works with

$$\text{finiteKL}(p, q) = \sum_a q_a \varphi\left(\frac{p_a}{q_a}\right), \quad \varphi(x) = x \log x + 1 - x,$$

for probability mass functions  $p, q$  on a finite type, where  $\varphi$  is mathlib's `klFun`. Lean's real division and logarithm are total, so on an atom with  $q_a = 0 < p_a$  this expression contributes 0 rather than  $+\infty$ . It is therefore *not* the extended KL divergence on arbitrary pairs. Every theorem below that is about KL carries the support hypothesis  $q_a = 0 \Rightarrow p_a = 0$ , under which the identification is proved.

| Lean name                                                           | Statement                                                                                                                                                                                                |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>finiteKL_eq_sum_log_of_support</code>                         | Under $\text{supp } p \subseteq \text{supp } q$ : $\text{finiteKL}(p, q) = \sum_a p_a \log(p_a/q_a)$ , i.e. ordinary KL.                                                                                 |
| <code>finiteKL_nonneg</code><br><code>finiteKL_self</code>          | $\text{finiteKL} \geq 0$ and $\text{finiteKL}(p, p) = 0$ .                                                                                                                                               |
| <code>finiteKL_map_le_of_support</code>                             | For any map $f$ between finite types and $\text{supp } p \subseteq \text{supp } q$ : $\text{finiteKL}(f_*p, f_*q) \leq \text{finiteKL}(p, q)$ . No surjectivity is required.                             |
| <code>finiteKL_map_le_of_pos</code><br><code>finiteKL_map_le</code> | Specializations to a strictly positive reference law (the second keeps a surjectivity argument that the proof does not use).                                                                             |
| <code>finiteKL_reversal_map_le_of_support</code>                    | If observation $f$ commutes with involutions $r, s$ ( $f \circ r = s \circ f$ ) and $\text{supp } p \subseteq \text{supp } r_*p$ , then $\text{finiteKL}(f_*p, s_*f_*p) \leq \text{finiteKL}(p, r_*p)$ . |
| <code>finiteKL_path_observe_le</code>                               | The path case: laws on $\text{Fin } n \rightarrow \alpha$ , reversal $i \mapsto \text{rev } i$ , pointwise observation; sparse reverse support is allowed.                                               |
| <code>reversePath_involutive</code>                                 | Path reversal is an involution.                                                                                                                                                                          |

Together these give Theorem 5. The proof applies Jensen’s inequality fibre by fibre: positive fibre mass permits normalized weights, atoms with zero reference mass have zero first-law mass by hypothesis, and empty fibres contribute zero. The commutation hypothesis in the reversal theorem is stated for arbitrary maps  $r, s$ ; the involution property is used only to interpret  $r_*p$  as the reversed law.

## B.2 Gibbs duality (`Duality.lean`)

For a nonempty finite type, a cost  $u$ , and a real price  $\lambda$ , the development defines entropy, mean cost, the partition function  $Z(\lambda) = \sum_a e^{-\lambda u_a}$ , and the Gibbs law. The theorems about a general  $q$  share two hypotheses:  $q$  is strictly positive, and  $\log q_a = -\lambda u_a - \log Z$  for some constant  $\log Z$  (Gibbs recognition).

| Lean name                                                                                                                                         | Statement                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>gibbsPMF</code><br><code>gibbsPMF_toReal</code><br><code>gibbsPMF_pos</code><br><code>gibbsPMF_log</code><br><code>gibbs_KL_identity</code> | The Gibbs law $e^{-\lambda u_a}/Z(\lambda)$ is constructed as a probability mass function, with normalization proved rather than assumed; it is strictly positive, and $\log q_a = -\lambda u_a - \log Z$ .                                                                                                                                              |
| <code>gibbs_penalized_optimal</code><br><code>gibbs_budget_optimal</code>                                                                         | For a strictly positive $q$ with $\log q_a = -\lambda u_a - \log Z$ : $\text{finiteKL}(p, q) = -H(p) + \lambda C(p) + \log Z$ (Theorem 3(i), one row).<br>$H(p) - \lambda C(p) \leq H(q) - \lambda C(q)$ (Theorem 3(ii)).<br>If $\lambda \geq 0$ , $C(p) \leq b$ , $C(q) \leq b$ , and $\lambda(C(q) - b) = 0$ , then $H(p) \leq H(q)$ (Theorem 3(iii)). |
| <code>gibbs_budget_shadow_bound</code>                                                                                                            | If $\lambda \geq 0$ , $C(p) \leq b'$ , and $\lambda(C(q) - b) = 0$ , then $H(p) \leq H(q) + \lambda(b' - b)$ (the inequality in Theorem 3(iv)).                                                                                                                                                                                                          |
| <code>maxent_budget_optimal</code>                                                                                                                | The same optimality conclusion for the constructed <code>gibbsPMF</code> : the Gibbs hypotheses are discharged, and $\lambda \geq 0$ , feasibility, and complementary slackness remain as hypotheses.                                                                                                                                                    |

## B.3 Scope

What is mechanized is finite supported KL data processing and its path form, together with single-row Gibbs duality. The following are proved on paper in Appendix A but not mechanized: the weighted-row extension, the price regimes and derivatives of Proposition 4, the reversal-mixture proposition, the packaging bound, and the ladder ranks. The extended-real form of data

processing for pairs with infinite KL is standard but also not mechanized. Resource budgets enter the Lean statements only as hypotheses. Nothing in the development derives a budget from packaging or persistence, and nothing verifies the stochastic experiments.

## C Evidence and reproducibility

Code, experiments, and the Lean development are in the public repository <https://github.com/ioannist/six-birds-currency>. Every numerical result in Section 5 is read from the receipts under `review/`, with one exception: the drive-parameter ranks of the ladder, which `scripts/make_paper_figures.py` computes directly (checking the fine-level value against the receipt) and writes to `paper/figures/ladder_family_rank.csv`. The same script draws every figure from these files. Experimental settings are the default parameters of the experiment scripts.

**Regenerating the evidence.** In an environment with the Python dependencies declared in `pyproject.toml`,

```
PYTHONPATH=src MPLBACKEND=Agg python3 scripts/math_review_checks.py
PYTHONPATH=src:scripts python3 scripts/make_paper_figures.py
PYTHONPATH=src python3 -m pytest -q
cd lean && lake build && lake env lean CheckAxioms.lean
```

The first command reruns all five experiments with their fixed designs. It writes one summary table per experiment and a receipt, `numerical_receipt.json`, that records the commands, the SHA-256 hashes of the selected Python and Lean source files, and the headline values. The test suite (63 tests) includes regression tests for each mathematical edge case discussed in the paper: infinite empirical audits, singular path mass, minimum-cost and slack budgets, infinite likelihood maximizers, cycle-rank counterexamples, and the packaging bound. `review/validation_receipt.json` records a passing run of the tests, the linter, the Lean build, and the axiom check.

**Table 2:** Where each reported result comes from. Files are under `review/` unless a path is given.

| Result                                       | Source                                                                                                     | Script                                                                                  |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Audit margins, missing-reverse mass (Fig. 2) | <code>dpi_checked.csv</code>                                                                               | <code>scripts/exp_dpi_scan.py</code>                                                    |
| Price curve (Fig. 3)                         | <code>budget_checked.csv</code>                                                                            | <code>scripts/exp_budget_sweep.py</code>                                                |
| Cycle ranks and affinity norms (Fig. 4)      | <code>ladder_checked.csv</code>                                                                            | <code>scripts/exp_currency_ladder.py</code>                                             |
| Drive-parameter ranks (Fig. 4a)              | <code>paper/figures/ladder_family_rank.csv</code> ; fine level also in <code>numerical_receipt.json</code> | <code>scripts/make_paper_figures.py</code> ; <code>scripts/math_review_checks.py</code> |
| Path-to-triangle counterexample              | <code>numerical_receipt.json</code>                                                                        | <code>scripts/math_review_checks.py</code>                                              |
| Held-out likelihoods and prices (Fig. 5)     | <code>proxy_checked.csv</code>                                                                             | <code>scripts/exp_proxy_ablation.py</code>                                              |
| Packaging defects, bounds, spending (Fig. 6) | <code>idempotence_checked.csv</code>                                                                       | <code>scripts/exp_idempotence_vs_budget.py</code>                                       |
| Lean build and axioms (App. B)               | <code>validation_receipt.json</code> , <code>lean_axioms.txt</code>                                        | <code>lean/CheckAxioms.lean</code>                                                      |

**Design choices.** All experiments use fixed, prespecified designs and report every outcome. The path-audit experiment draws one sample of fixed size per seed. The proxy experiment uses the requested design seed and a single proxy permutation for all samples, and records its

predeclared criteria whether or not they hold. The ladder construction is deterministic, so its seed labels are repetitions. Numerical roots are floating-point approximations; each root of the budget solver is reported with the cost it actually achieves. Stationary laws are computed by an eigenvector method in the ladder and packaging experiments, and by lazified power iteration elsewhere (including stationary-conditional lumping when no stationary law is supplied); either way they are accepted only after a stationarity-residual check. None of the computations is an exact or interval-arithmetic certificate.

## D Version history

**v2 (October 1, 2026; DOI 10.5281/zenodo.23088986).** The currency–constraint principle is stated as a conditional theorem, with the resource law as an explicit premise and a counterexample showing that persistence alone does not supply one. The finite Gibbs duality is proved in full, with all price regimes, and its single-row form is mechanized in Lean. The data-processing theorem is extended to arbitrary deterministic maps and sparse supports, with a path-reversal form. Empirical path audits treat unmatched reversals as infinite asymmetry and use a reversal-mixture regularization that satisfies data processing on the same sample. The proxy experiment uses a fixed design and a fixed proxy. The manuscript uses held-out prediction as the supported comparison and no longer draws a price-stability conclusion: across-sample price stability, reported as a comparison in v1, is not supported under the fixed design. The experiment still records the price dispersions and the outcome of its original criterion. The ladder result separates available cycle dimension from driven dimension. A uniform packaging-defect bound, and the distinction between calibration cost and actual spending, were added. Figures and the exposition were rewritten throughout.

**v1 (March 9, 2026; DOI 10.5281/zenodo.18926771).** First public version.

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